

RPi Camera LED interposer: sits between a Raspberry Pi and its camera.
 MIPI lanes, I2C, GND and 3V3 pass straight through J1 -> J2.
 U1 (I2C slave on the camera bus) drives the camera GPIO pins 11/12,
 PWM-drives the illumination LEDs and reads ambient light on its ADC.

LED current is set by R3/R4 against the 3V3 rail (about 30–40mA each with $V_f=2.9V$). Total draw comes from the Pi camera 3V3 supply – keep the duty/current within the Pi's camera connector budget.

